Accounting Module — High-Level Design (HLD) Architecture

> ***Project:*** *GTR Service ERP Platform*
>
> ***Module:*** *Accounts (`ACCOUNTS` context)*
>
> ***Stack:*** *Java 21 · Spring Boot 3.5 · React + Vite · MySQL · Liquibase · JWT*
>
> ***Pattern:*** *Multi-Tenant SaaS (Database-per-Tenant)*
>
> ***Liquibase Context Tag:*** *`ACCOUNTS`*

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1. Module Overview

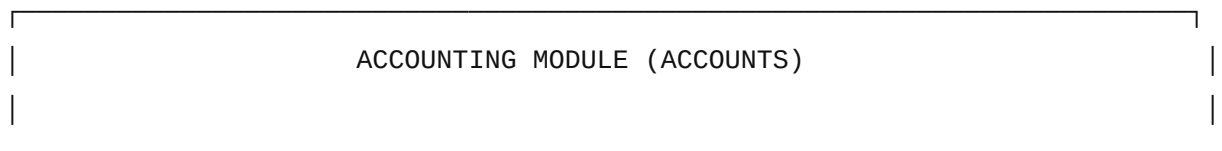
The **Accounting Module** is the financial backbone of the GTR Service ERP platform. It implements a full **double-entry bookkeeping** system compliant with **IFRS** standards and **UAE VAT (5%)** regulations.

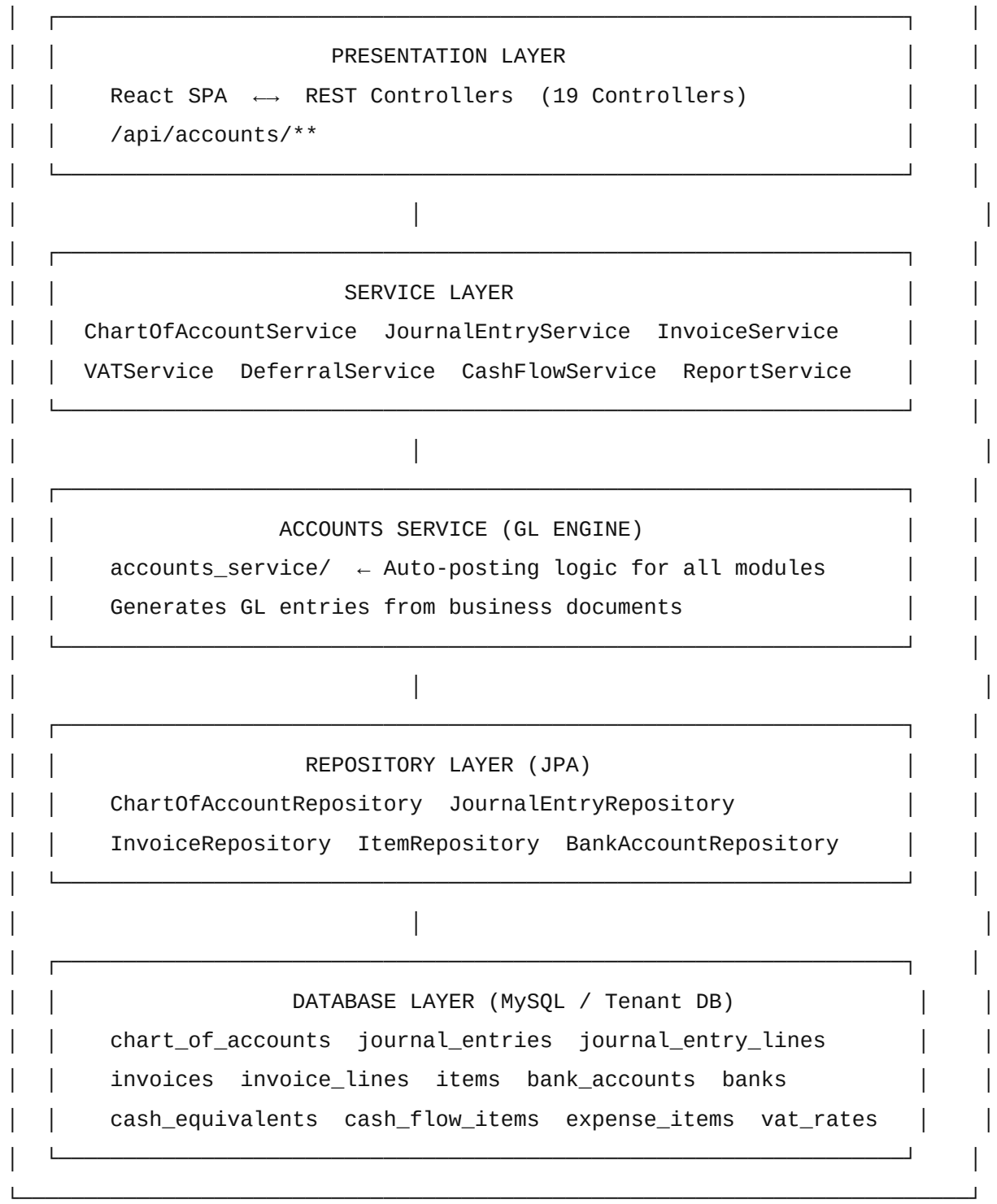
What It Does

Capability	Description
Chart of Accounts	Hierarchical COA with Balance Sheet & P&L sections
General Ledger (GL)	All financial transactions recorded as Dr/Cr journal entries
Accounts Payable (AP)	Supplier invoices, payments, PDC management
Accounts Receivable (AR)	Customer invoices, receipts, credit notes
VAT Compliance	UAE VAT 5%, Input/Output VAT, VAT Return file
Deferred Accounting	Deferred expenses & revenue recognition (monthly/daily/manual)
Financial Statements	Balance Sheet, P&L, Cash Flow, Trial Balance
Bank Reconciliation	Bank accounts, cash on hand, cheque tracking
Warehouse Accounting	Inventory transfer, write-off, receipt GL entries
Opening Balances	Customer, supplier, employee salary opening balances

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2. Core Architecture — Layer Diagram





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3. Sub-Modules & Functional Breakdown

```
ACCOUNTS Module
│
├──    Chart of Accounts (COA)
│       ├── Account Hierarchy (Group → Account)
│       ├── Balance Sheet Sections (Assets, Liabilities, Equity)
│       ├── Income Statement Sections (Income, Expense)
│       ├── Dimensions (Item, Warehouse, Department)
```

```
|       └── Off-Balance Sheet Accounts
|
├──    General Ledger (GL Engine)
|       ├── Journal Entries (Manual + Auto-Posted)
|       ├── Journal Entry Lines (Dr / Cr)
|       ├── Audit Log
|       └── Account Statement
|
├──    Accounts Payable (AP)
|       ├── Supplier Invoices (Invoice Received)
|       ├── Debit Notes / Invoice Corrections
|       ├── Purchase Payments (Bank / Cash / PDC)
|       ├── Prepayments to Suppliers
|       ├── Return of Goods (Supplier)
|       └── Mutual Settlement
|
├──    Accounts Receivable (AR)
|       ├── Customer Invoices (Sales Invoice)
|       ├── Credit Notes / Invoice Corrections
|       ├── Receipts (Bank / Cash / PDC)
|       ├── Prepayments from Customers
|       ├── Trade Discounts & Rebates
|       └── Mutual Settlement
|
├──    Cash & Bank Management
|       ├── Bank Accounts
|       ├── Cash on Hand
|       ├── PDC Issued / Received (Post-Dated Cheques)
|       ├── Cash Equivalents
|       └── Cash Flow Transactions
|
├──    VAT / Tax Engine
|       ├── VAT Groups & Types
|       ├── VAT Rates (0%, 5%, Exempt, Out-of-Scope)
|       ├── Input VAT / Output VAT
|       ├── Tax Transactions (Manual adjustments)
|       ├── VAT Return File (UAE FTA)
|       └── VAT Liability Calculation & Payment
|
├──    Deferred Accounting
|       ├── Deferred Expenses (from Purchase Invoices)
|       ├── Expense Recognition (Monthly / Daily / Manual)
|       ├── Deferred Revenue (from Customer Invoices)
```

```
│       ├── Revenue Recognition Schedules
│       └── Deferral Reports & Adjustments
│
├──    Company & Currency Setup
│       ├── Company Master
│       ├── Company GL Accounts (Default AP/AR/Bank accounts)
│       ├── Currency Master
│       └── Equity Accounts
│
├──    Item GL Accounts (Integration Bridge)
│       ├── Default GL Accounts per Item Type
│       ├── Resource Types: Inventory / Fixed Asset / Manpower / Service
│       └── Item-Level Overrides
│
└──    Financial Reporting
        ├── Trial Balance (with Turnovers)
        ├── General Ledger Report
        ├── Income Statement (P&L)
        ├── Balance Sheet (Statement of Financial Position)
        ├── Cash Flow Statement
        ├── Statement of Account (per GL Account / per Customer / per Supplier)
        ├── VAT Output / Input Transaction Reports
        ├── Deferred Expense Balance Report
        ├── Deferred Revenue Report
        └── Posting Reports (per Document)
```

---

4. Chart of Accounts (COA) Engine

4.1 Account Hierarchy

```
COA (Chart of Accounts)
│
├── BALANCE SHEET (BS)
│   ├── Assets
│   │   ├── BS Inventories                ← Goods, Raw Materials, WIP, Finished Goods
│   │   ├── BS Fixed Assets (PPE)         ← Property, Plant & Equipment
│   │   ├── BS Intangible Assets
│   │   ├── BS Investment Property
│   │   ├── BS Current Financial Assets
│   │   ├── BS Cash Equivalents           ← Short-term liquid investments
```

```
│   │   ├── BS Deferred Expenses           ← Prepaid costs being amortized
│   │   ├── BS Deferred Tax Assets
│   │   ├── BS Current Income Tax Receivable
│   │   ├── BS Current VAT Receivable      ← Input VAT recoverable
│   │   ├── BS Trade Receivables           ← Customer balances
│   │   ├── BS Prepayments to Suppliers
│   │   └── BS Assets Classified as Held for Sale
│   │
│   ├── Liabilities
│   │   ├── BS Current Trade Payables      ← Supplier balances
│   │   ├── BS Prepayments from Customers
│   │   ├── BS Current VAT Payable         ← Output VAT owed to FTA
│   │   ├── BS Deferred Revenue            ← Revenue received but not yet earned
│   │   ├── BS Deferred Tax Liabilities
│   │   ├── BS Current Income Tax Payable
│   │   ├── BS PDC Issued to Companies     ← Cheques issued, not yet cleared
│   │   ├── BS PDC Received from Companies ← Cheques received, not yet deposited
│   │   └── BS Current Financial Liabilities
│   │
│   └── Equity
│       ├── Owner's Equity
│       ├── Retained Earnings
│       └── BS Non-Controlling Interests
│
└── INCOME STATEMENT (P&L)
    ├── Income
    │   ├── Sales of Goods
    │   ├── Sales of Services
    │   ├── Other Operating Income
    │   └── Trade Discounts and Rebates on Sales (contra)
    │
    └── Expenses
        ├── COS: Goods for Resale          ← Cost of goods sold
        ├── COS: Goods for Production
        ├── Input VAT on Purchases
        ├── Other Administrative Expenses
        └── Other Operating Expenses
```

4.2 Account Attributes (Entity: `ChartOfAccount`)

Attribute	Purpose
code	Unique GL account code (e.g., 2010400)

Attribute	Purpose
`name`	Account name
`type`	Debit / Credit / Debit-Credit
`section`	BS section or P&L section (drives financial statement placement)
`parentGroup`	COA hierarchy parent
`isQuantitative`	If true, tracks both amount AND quantity
`offBalance`	Off-balance sheet flag
`dimensions`	Analytical dimensions: Item, Warehouse, Department

4.3 COA Dimensions

```
Dimensions (AccountDimension)
    ├── ITEM       ← Track balance per product/SKU
    ├── WAREHOUSE  ← Track balance per warehouse location
    └── DEPARTMENT ← Track balance per cost center/department
```

> ***Why dimensions matter:*** *A goods account with ITEM + WAREHOUSE dimensions lets the Trial Balance show: "Goods Account — Item: Wire Coil — Warehouse: Main Store — Balance: 500 Kg / AED 12,500"*

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5. GL Transaction Engine (Double-Entry Bookkeeping)

5.1 Core Principle

Every financial event creates a **Journal Entry** with balanced Dr/Cr lines:

```
JournalEntry
    ├── entryNumber (auto)
    ├── entryDate
    ├── description
```

```
    ├── status: DRAFT | POSTED | REVERSED
    ├── documentType: INVOICE | PAYMENT | JOURNAL | VAT | etc.
    ├── referenceDocument (FK to source: invoice_id, payment_id, etc.)
    └── JournalEntryLines[]
            ├── glAccount (FK → chart_of_accounts)
            ├── debitAmount
            ├── creditAmount
            ├── dimension1 (item_id)
            ├── dimension2 (warehouse_id)
            └── description
```

5.2 Auto-Posting Rules

The `accounts_service` layer contains the **GL posting engine** — it auto-generates journal entries when business documents are posted:

```
Business Document Posted
        │
        ▼
accounts_service.postDocument(document)
        │
        ├── Determines document type
        ├── Looks up Item GL Accounts (ItemGLAccount)
        ├── Looks up Company Default GL Accounts (CompanyGLAccount)
        ├── Applies VAT rules (VATService)
        ├── Builds JournalEntry + JournalEntryLines
        └── Saves to journal_entries (status = POSTED)
```

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6. Purchases Sub-Ledger

6.1 Purchase Invoice Received — GL Mapping

When a **Purchase Invoice** is received and posted, the system auto-generates:

```
┌──────────────────────────────────────────────────────────┐
│                  INVOICE RECEIVED (AP)                   │
├──────────────────┬───────────────────────────────────────┤
│  Debit (Dr)      │  Credit (Cr)                          │
├──────────────────┼───────────────────────────────────────┤
│ Goods (Inventory │ Current Trade Payables (Supplier balance)  │
│  account)        │                                       │
```

Input VAT	Current Trade Payables (VAT portion)
Admin Expenses	Current Trade Payables (Expense portion)

6.2 Purchase Payment — GL Mapping

Payment Method	Dr	Cr
Bank Transfer	Current Trade Payables	Bank Account
Cash	Current Trade Payables	Cash on Hand
Cheque (PDC)	Current Trade Payables	PDCs Issued to Companies

> ***Bank charges on PDC:*** *Additional entry Dr Bank Charges / Cr Bank Account*

6.3 Supplier Prepayment

PREPAYMENT TO SUPPLIER	
Dr	Cr
Prepayments to Suppliers	Bank / Cash on Hand / PDCs Issued to Companies

6.4 Return of Goods to Supplier

RETURN OF GOODS (SUPPLIER)	
Dr	Cr
Prepayments to Suppliers	Goods (Inventory)
	COS: Goods for Resales
	Current VAT Payable
Current Trade Payables	Prepayment from Customer

6.5 Debit Note / Invoice Correction

Event	Dr	Cr
Debit Note	Goods	Prepayments to Suppliers
	Prepayment to Supplier	Current Trade Payables
Mutual Settlement	Current Trade Payables	Prepayment to Supplier

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7. Sales Sub-Ledger

7.1 Customer Invoice — GL Mapping

CUSTOMER INVOICE (AR)	
Debit (Dr)	**Credit (Cr)**
Trade Receivables (full invoice)	Sales of Goods
Trade Receivables (VAT portion)	Current VAT Payable (Output VAT)
Trade Discount & Rebates	Sales of Goods (contra for discount)

7.2 Payment Receipt — GL Mapping

Payment Method	Dr	Cr
Bank Transfer	Bank Account	Current Trade Receivables
Cash	Cash on Hand	Current Trade Receivables
Cheque (PDC)	PDCs Received from Companies	Current Trade Receivables

7.3 Customer Prepayment

```
Prepayment from Customer:
```

```
  Dr: Cash / Bank / PDC Received
  Cr: Prepayment from Customer (Liability account)
```

7.4 Credit Note / Invoice Correction

CREDIT NOTE (AR — CUSTOMER)	
Dr	Cr
Prepayment from Customer	Sales of Goods (reversal)
	Current VAT Payables (reversal)
Sales of Goods (Return of Goods)	Prepayment from Customer

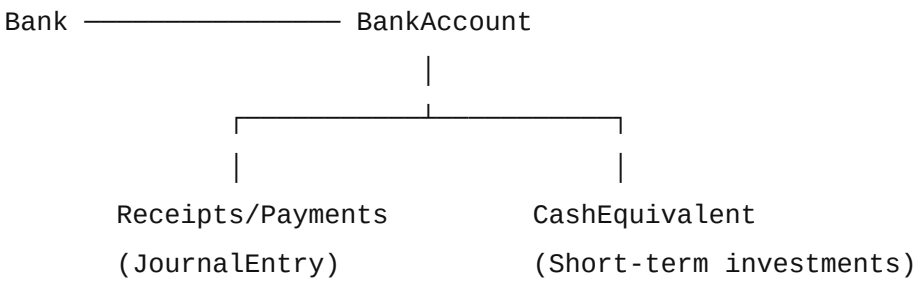
7.5 Mutual Settlement (AP/AR Netting)

```
Mutual Settlement:
  Dr: Current Trade Payables (or Receivables)
  Cr: Current Trade Receivables (or Payables)
  (Net off the balances between customer/supplier who is both)
```

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8. Payments & Cash Management

8.1 Entity Relationships



8.2 Cash Flow Tracking

```
CashFlowItem
    ├── OPERATING  ← Customer receipts, supplier payments
    ├── INVESTING  ← Asset purchases, investments
```

```
    └── FINANCING  ← Loans, equity injections

CashFlowTransaction
    ├── cashFlowItem (FK)
    ├── amount
    ├── transactionDate
    └── description
```

8.3 PDC (Post-Dated Cheque) Management

```
PDC Received (from Customer):
    Step 1 — Receive cheque:
        Dr: PDCs Received from Companies
        Cr: Current Trade Receivables

    Step 2 — On clearing date:
        Dr: Bank Account
        Cr: PDCs Received from Companies

PDC Issued (to Supplier):
    Step 1 — Issue cheque:
        Dr: Current Trade Payables
        Cr: PDCs Issued to Companies
        + Record: cheque date, cheque number, bank charges

    Step 2 — On clearing date:
        Dr: PDCs Issued to Companies
        Cr: Bank Account
```

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9. VAT / Tax Module

9.1 VAT Architecture

```
VAT Engine
│
├── VAT Groups
│       └── Group multiple VAT types (e.g., Standard, Exempt)
│
├── VAT Types
│       ├── Current TAX Receivable → GL: Current VAT Receivable (Input VAT)
│       └── Current TAX Payable    → GL: Current VAT Payables (Output VAT)
```

```
│
├── VAT Rates
│       ├── 0%        ← Zero-rated supplies
│       ├── 5%        ← Standard UAE VAT
│       ├── Exempt    ← Exempted supplies
│       └── Out of Scope ← Not subject to VAT
│
├── Out-of-Scope Reasons
│       ├── Intra-company Transaction
│       ├── Company not registered for VAT
│       └── Product not subject to VAT
│
└── Tax Transactions (Manual Adjustments)
        ├── Types: Tax / Penalties / Interest / Tax Prepayment
        └── Transaction Subtypes:
                ├── Payable Tax Increase   ← Increase what you owe FTA
                ├── Payable Tax Decrease   ← Decrease what you owe FTA
                ├── Recoverable Tax Increase ← Increase Input VAT claim
                └── Recoverable Tax Decrease ← Decrease Input VAT claim
```

9.2 VAT Transaction Flow

```
Purchase Invoice Posted
    │
    ▼
Dr: Goods Account          (net amount)
Dr: Current VAT Receivable (Input VAT = 5% of net)
    Cr: Current Trade Payables (gross = net + VAT)

Sales Invoice Posted
    │
    ▼
Dr: Trade Receivables      (gross = net + VAT)
    Cr: Sales of Goods     (net amount)
    Cr: Current VAT Payable (Output VAT = 5% of net)
```

9.3 VAT Liability Calculation & Settlement

```
VAT Liability = Output VAT (Payable) − Input VAT (Receivable)

VAT Return Period:
    1. Calculate VAT Liability
    2. Generate VAT Return File (UAE FTA format)
```

```
    3. Make payment to FTA:
        Dr: Current VAT Payable
        Cr: Bank Account
```

---

10. Deferred Expenses & Revenue Recognition

10.1 Deferred Expenses

```
Purchase Invoice received with deferred expense item
    |
    ▼
System creates Deferred Expense Record:
    Dr: Deferred Expenses (BS Asset)
    Cr: Current Trade Payables

Expense Recognition runs (Monthly / Daily / Manual):
    |
    ▼
Per recognition period:
    Dr: Other Administrative Expenses (P&L)
    Cr: Deferred Expenses (BS Asset)

Until the full deferred amount is recognized (BS balance = 0).
```

10.2 Deferred Revenue

```
Customer Invoice raised for future service delivery
    |
    ▼
System creates Deferred Revenue Record:
    Dr: Trade Receivables (or Cash if prepaid)
    Cr: Deferred Revenue (BS Liability)

Revenue Recognition runs (Monthly / Daily / Manual):
    |
    ▼
Per recognition period:
    Dr: Deferred Revenue (BS Liability)
    Cr: Sales of Services (P&L Income)

Until full revenue is earned (BS liability = 0).
```

10.3 Recognition Methods

Method	How It Works
Monthly	Total / Number of months = equal monthly amount
Daily	Total / Number of days = daily proration
Manual	User enters each recognition amount manually

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11. Financial Reporting Engine

11.1 Report Hierarchy

```
Financial Reports
│
├──    Transactional Reports
│       ├── Posting Report      ← Per document: shows all GL lines generated
│       ├── General Ledger      ← All movements on selected GL accounts, period-wise
│       └── Statement of Account ← Movements for a specific GL account/partner
│
├──    Analytical Reports
│       ├── Trial Balance          ← Opening balance + turnovers + closing balance
│       ├── Trial Balance Turnovers ← Detailed Dr/Cr movements per account
│       └── Work in Progress Balances (for WIP GL accounts)
│
├──    Financial Statements (IFRS)
│       ├── Income Statement (P&L)          ← Revenues - Costs - Expenses = Net
Profit
│       ├── Balance Sheet                   ← Assets = Liabilities + Equity
│       ├── Cash Flow Statement             ← Operating + Investing + Financing
│       └── Statement of Changes in Equity  ← Equity movements
│
├──    VAT Reports
│       ├── VAT Output Transaction Report   ← All output VAT (sales)
│       ├── VAT Input Transaction Report    ← All input VAT (purchases)
│       ├── VAT Return File                 ← UAE FTA submission format
│       └── Statement of Tax Account        ← VAT ledger view
│
└──    Deferral Reports
```

```
        ├── Deferred Expense Balance Report ← Remaining unrecognized expenses
        └── Deferred Revenue Report         ← Remaining unrecognized revenue
```

11.2 Trial Balance Logic

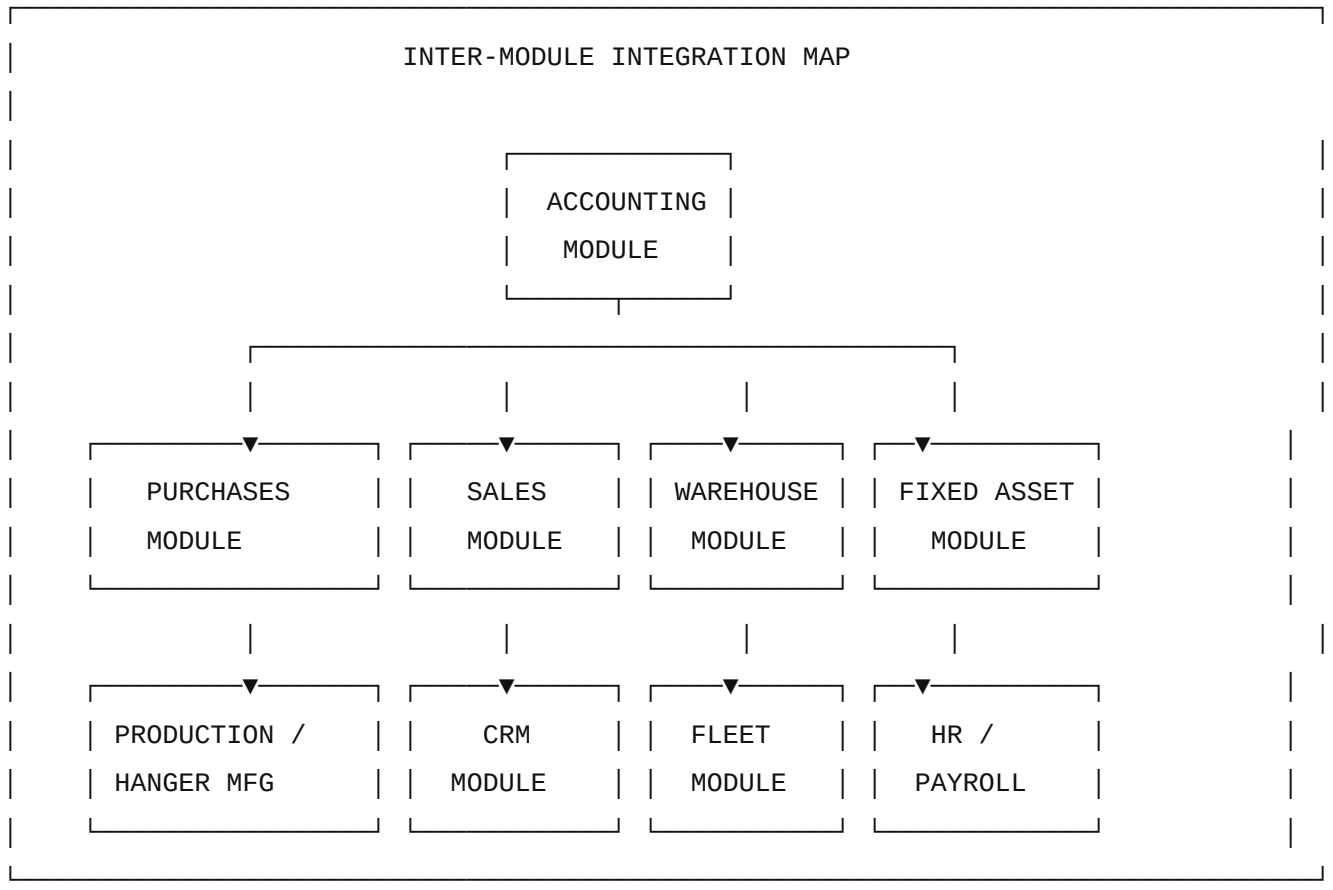
```
For each GL Account:
    Opening Balance   = Sum of all Dr-Cr before the selected period
    Debit Turnover    = Sum of Dr entries IN the period
    Credit Turnover   = Sum of Cr entries IN the period
    Closing Balance   = Opening + Debit Turnover - Credit Turnover

If account has dimensions (Item, Warehouse):
    → Trial Balance can drill down by dimension
    → Example: Goods Account → by Warehouse → by Item
```

---

12. Inter-Module Integration Map

This is the most critical section — it shows exactly how **Accounting connects with every other module** in the system.



12.1 Integration: Accounting ↔ Purchases Module

Event in Purchases	GL Entry Generated in Accounts
Purchase Order **Received** (GRN)	Dr Goods / Cr Goods-in-Transit
Invoice Received posted	Dr Goods + Dr Input VAT / Cr Trade Payables
Payment Made (Bank)	Dr Trade Payables / Cr Bank Account
Payment Made (Cash)	Dr Trade Payables / Cr Cash on Hand
Payment Made (Cheque/PDC)	Dr Trade Payables / Cr PDC Issued to Companies
Supplier Return posted	Reversal of original invoice GL entries
Debit Note issued	Dr Goods / Cr Prepayment to Suppliers
Prepayment to Supplier	Dr Prepayments to Suppliers / Cr Bank/Cash
Mutual Settlement	Dr Trade Payables / Cr Prepayments to Suppliers

Integration Point:

```
// In PurchasesService → after invoice.setStatus(POSTED):
accountsService.postPurchaseInvoice(invoiceId, tenantId);
// → Creates JournalEntry + Lines automatically
```

12.2 Integration: Accounting ↔ Sales Module

Event in Sales	GL Entry Generated in Accounts
Customer Invoice posted	Dr Trade Receivables / Cr Sales of Goods + Cr VAT Payable
Trade Discount applied	Dr Trade Discount & Rebates / Cr Sales of Goods
Receipt — Bank	Dr Bank Account / Cr Trade Receivables

Event in Sales	GL Entry Generated in Accounts
Receipt — Cash	Dr Cash on Hand / Cr Trade Receivables
Receipt — Cheque/PDC	Dr PDC Received / Cr Trade Receivables
Credit Note posted	Dr Sales of Goods (reversal) / Cr Trade Receivables
Customer Return posted	Complex reversal (see Sales sub-ledger section)
Prepayment from Customer	Dr Bank/Cash / Cr Prepayment from Customer (Liability)
Mutual Settlement	Dr Prepayment from Customer / Cr Trade Receivables
Deferred Revenue item invoiced	Dr Trade Receivables / Cr Deferred Revenue

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12.3 Integration: Accounting ↔ Warehouse Module

Event in Warehouse	GL Entry Generated in Accounts
Inventory Transfer (warehouse to warehouse)	Dr Goods (Destination WH) / Cr Goods (Source WH)
Inventory Write-Off	Dr Other Operating Expenses / Cr Goods
Inventory Receipt (physical count gain)	Dr Goods / Cr Other Operating Income
Stock Adjustment	Dr/Cr respective Goods GL account

Dimension tracking: When Goods GL Account has WAREHOUSE dimension, all inventory movements are tracked per warehouse — the Trial Balance can show "Goods in Main WH = AED 50,000 vs Goods in Branch WH = AED 12,000"

12.4 Integration: Accounting ↔ Fixed Assets Module

Event in Fixed Assets	GL Entry Generated in Accounts
Asset Acquisition posted	Dr Fixed Assets (PPE) / Cr Bank/Trade Payables
Depreciation run	Dr Depreciation Expense / Cr Accumulated Depreciation
Asset Disposal	Dr Accumulated Depreciation + Dr Loss on Disposal / Cr Fixed Asset + Cr Proceeds
Asset Revaluation	Dr Fixed Asset / Cr Revaluation Reserve (Equity)
Capital Improvement	Dr BS Capital Improvements to PPE / Cr Bank

---

12.5 Integration: Accounting ↔ Production / Hanger Manufacturing

Event in Production	GL Entry Generated in Accounts
Material Issue to production	Dr WIP (Work in Progress) / Cr Goods (Raw Materials)
Production Completed	Dr Finished Goods / Cr WIP
Scrap Recorded	Dr Scrap Expense / Cr WIP
Production Cost Allocation	Dr WIP / Cr Wages Payable or Overhead
Project Invoice (service)	Dr Trade Receivables / Cr Sales of Services + Cr VAT

12.6 Integration: Accounting ↔ HR / Payroll Module

Event in HR/Payroll	GL Entry Generated in Accounts
Payroll Run	Dr Salaries Expense / Cr Salary Payable

Event in HR/Payroll	GL Entry Generated in Accounts
Salary Payment	Dr Salary Payable / Cr Bank Account
Employee Advance	Dr Employee Advances / Cr Cash/Bank
End of Service (EOSB)	Dr EOSB Expense / Cr EOSB Provision
Opening Balances — Employee salary	Dr Salary Expense / Cr Salary Payable (opening entry)

12.7 Integration: Accounting ↔ CRM Module

Event in CRM	GL Entry Generated in Accounts
Rental Quotation → Invoice	Feeds into Sales sub-ledger (see 12.2)
Expense Booking from CRM activity	Dr Other Administrative Expenses / Cr Trade Payables

---

12.8 Integration: Accounting ↔ Fleet Module

Event in Fleet	GL Entry Generated in Accounts
Fuel Expense recorded	Dr Fuel Expense / Cr Cash/Bank/Trade Payables
Vehicle Maintenance	Dr Maintenance Expense / Cr Trade Payables
Trip Invoice	Feeds into Sales sub-ledger AR
Driver Advance	Dr Driver Advance / Cr Cash/Bank

---

13. API Surface & Controller Map

Controller	Base URL	Key Operations
`ChartOfAccountController`	`/api/accounts/coa`	CRUD, hierarchy, sections
`JournalEntryController`	`/api/accounts/journal`	Create, post, reverse, search
`InvoiceController`	`/api/accounts/invoices`	AP & AR invoice management
`ItemController`	`/api/accounts/items`	Item master + resource types
`ItemGLAccountController`	`/api/accounts/item-gl`	Default GL per item
`BankController`	`/api/accounts/banks`	Bank master
`BankAccountController`	`/api/accounts/bank-accounts`	Bank accounts per company
`CashEquivalentController`	`/api/accounts/cash-equiv`	Cash equivalents management
`CashFlowItemController`	`/api/accounts/cash-flow-items`	CF classification items
`CashFlowTransactionController`	`/api/accounts/cash-flow-tx`	Cash flow entries
`CashFlowStatementController`	`/api/accounts/cash-flow-stmt`	CF Statement report
`ExpenseItemController`	`/api/accounts/expense-items`	Expense item master

Controller	Base URL	Key Operations
ExpenseItemTypeController	/api/accounts/expense-types	Expense categories
IncomeItemController	/api/accounts/income-items	Income item master
FACashFlowItemController	/api/accounts/fa-cf-items	Fixed asset CF items
EquityStatementController	/api/accounts/equity-stmt	Equity statement report
CompanyController	/api/accounts/company	Company GL setup
CurrencyController	/api/accounts/currencies	Currency master
AuditLogController	/api/accounts/audit	GL audit trail

---

14. Database Schema Design

14.1 Core Tables

```
-- Chart of Accounts
chart_of_accounts (
    id, code, name, type,          -- DEBIT / CREDIT / DEBIT_CREDIT
    section,                       -- BS_INVENTORIES / BS_ASSETS / P&L_INCOME etc.
    parent_group_id,               -- Self-referencing for hierarchy
    is_quantitative,               -- Track quantities
    is_off_balance,
    description,
    tenant_id
)

-- GL Journal Entries
journal_entries (
    id, entry_number, entry_date,
```

```
    description, status,           -- DRAFT / POSTED / REVERSED
    document_type,                 -- INVOICE / PAYMENT / JOURNAL / VAT_SETTLEMENT
    reference_id,                  -- FK to source document
    reference_type,                -- "INVOICE" / "PURCHASE_PAYMENT" / "MANUAL"
    total_debit, total_credit,     -- Must always balance (Dr = Cr)
    created_by, created_at
)

-- GL Journal Entry Lines
journal_entry_lines (
    id, journal_entry_id,
    gl_account_id,                 -- FK → chart_of_accounts
    debit_amount, credit_amount,
    dimension_item_id,             -- Analytical dimension: Item
    dimension_warehouse_id,        -- Analytical dimension: Warehouse
    description
)

-- Invoices (AP + AR unified)
invoices (
    id, invoice_number, invoice_date, due_date,
    invoice_type,                  -- PURCHASE / SALES
    partner_id, partner_type,      -- SUPPLIER / CUSTOMER
    status,                        -- DRAFT / POSTED / PAID / PARTIALLY_PAID
    subtotal, vat_amount, total_amount,
    currency_id,
    journal_entry_id,             -- FK → auto-generated GL entry
    is_deferred, deferred_end_date
)

-- Invoice Lines
invoice_lines (
    id, invoice_id,
    item_id,                       -- FK → items
    description, quantity, unit_price,
    discount_percent, vat_rate,
    line_amount, vat_amount,
    gl_account_id                  -- Override GL account per line
)
```

14.2 VAT Tables

```
-- VAT Groups
```

```
vat_groups (id, name, description)

-- VAT Types
vat_types (
    id, name, vat_group_id,
    receivable_gl_account_id,     -- Current VAT Receivable
    payable_gl_account_id         -- Current VAT Payable
)

-- VAT Rates
vat_rates (
    id, name, rate,               -- 0.00 / 5.00
    vat_type,                     -- STANDARD / EXEMPT / OUT_OF_SCOPE / ZERO
    out_of_scope_reason           -- INTRA_COMPANY / NOT_REGISTERED / NOT_SUBJECT
)

-- Tax Transactions (manual adjustments)
tax_transactions (
    id, transaction_date,
    transaction_type,             -- PAYABLE_INCREASE / PAYABLE_DECREASE /
                                  -- RECOVERABLE_INCREASE / RECOVERABLE_DECREASE
    payment_type,                 -- TAX / PENALTIES / INTEREST / PREPAYMENT
    amount, description,
    journal_entry_id
)
```

14.3 Item & GL Mapping Tables

```
-- Item Master
items (
    id, code, name,
    item_type,                    -- INVENTORY / FIXED_ASSET / SERVICE / MANPOWER
    resource_type,                -- INVENTORY / FIXED_ASSET / MANPOWER / ALL /
SERVICE
    default_gl_account_id,
    is_deferred_eligible
)

-- Item-GL Account Default Mapping
item_gl_accounts (
    id, item_id,
    transaction_type,             -- SALES / PURCHASES / INVENTORY / COS / DEFERRED
    gl_account_id
```

```
)

-- Company Default GL Accounts
company_gl_accounts (
    id, company_id,
    account_purpose,              -- DEFAULT_AP / DEFAULT_AR / DEFAULT_BANK /
DEFAULT_CASH
    gl_account_id
)
```

14.4 Deferred Accounting Tables

```
-- Deferred Expenses
deferred_expenses (
    id, invoice_id, invoice_line_id,
    total_amount, recognized_amount, remaining_amount,
    start_date, end_date,
    recognition_method,           -- MONTHLY / DAILY / MANUAL
    status,                       -- ACTIVE / COMPLETED
    gl_deferred_account_id,
    gl_expense_account_id
)

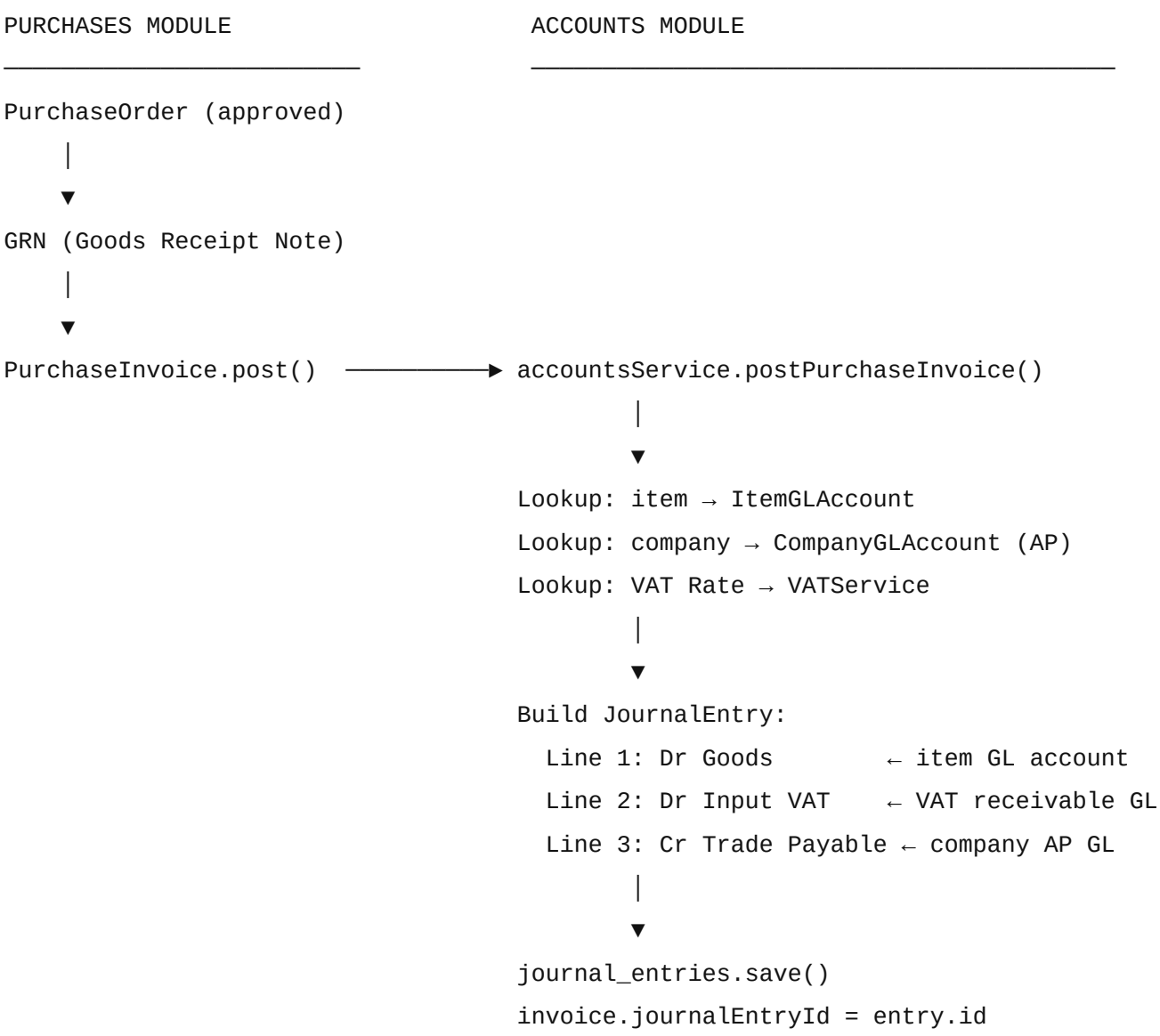
-- Deferred Revenue
deferred_revenues (
    id, invoice_id, invoice_line_id,
    total_amount, recognized_amount, remaining_amount,
    start_date, end_date,
    recognition_method,
    status,
    gl_deferred_account_id,       -- Deferred Revenue (Liability)
    gl_income_account_id          -- Sales of Services (Income)
)

-- Recognition Schedules
recognition_schedule (
    id, deferred_id, deferred_type,  -- EXPENSE / REVENUE
    recognition_date, amount,
    status,                           -- PENDING / RECOGNIZED
    journal_entry_id
)
```

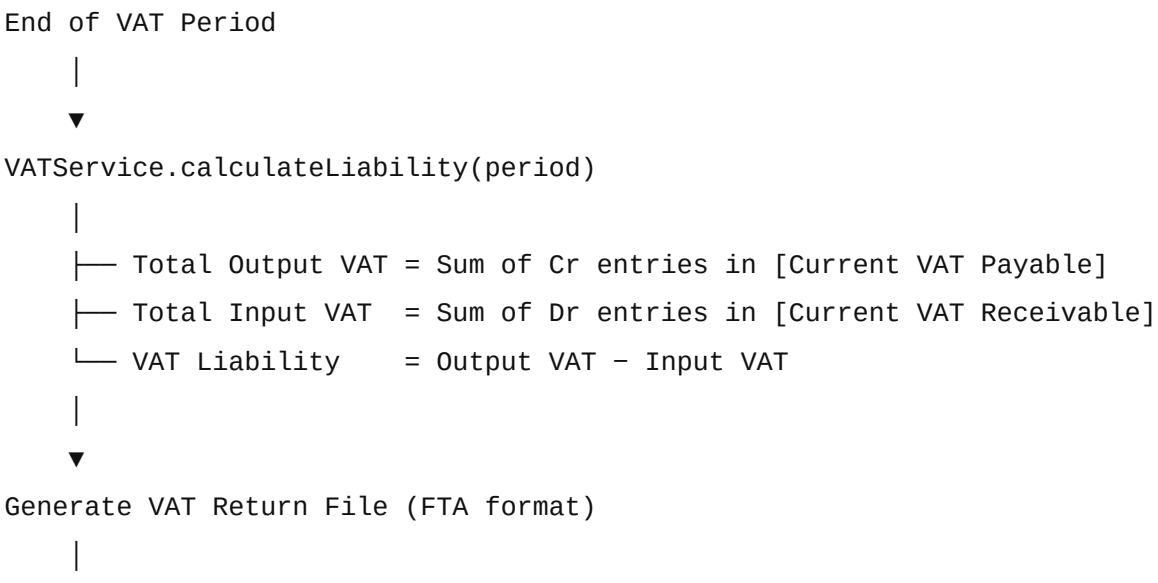
---

15. Data Flow Diagrams

15.1 Purchase Invoice → GL Flow



15.2 VAT Settlement Flow



```
    ▼
VATService.payVATLiability(bankAccountId, amount)
    │
    ▼
JournalEntry:
    Dr: Current VAT Payable     (clears the liability)
    Cr: Bank Account            (payment out)
```

15.3 Opening Balances Entry

```
Onboarding / System Go-Live
    │
    ▼
Manual Journal Entry (DocumentType = OPENING_BALANCE)
    │
    ├── Customer Opening Balances:
    │       Dr: Trade Receivables (Customer-wise if dimensions used)
    │       Cr: Opening Balance Equity Account
    │
    ├── Supplier Opening Balances:
    │       Dr: Opening Balance Equity Account
    │       Cr: Current Trade Payables
    │
    └── Employee Salary Opening Balances:
            Dr: Salary Expense
            Cr: Salary Payable
```

---

16. Module Activation (Multi-Tenant)

16.1 How the Accounting Module is Activated for a Tenant

```
Tenant Provisioning Request:
  modules: [ACCOUNTS, PURCHASES, SALES, WAREHOUSE, ...]
                │
                ▼
TenantRegistry.addOrUpdateTenant(tenant)
                │
                ▼
TenantSchemaCreator.ensureSchema(dataSource, tenant)
                │
                ▼
```

```
LiquibaseService.runUpdate(contexts = "ACCOUNTS,PURCHASES,SALES,...")
                │
                ▼
Only ACCOUNTS-tagged changesets run:
    ├── 060-create-accounts-coa.xml      context="ACCOUNTS"
    ├── 061-create-journal-entries.xml   context="ACCOUNTS"
    ├── 062-create-invoices.xml          context="ACCOUNTS"
    ├── 063-create-vat-tables.xml        context="ACCOUNTS"
    ├── 064-create-bank-accounts.xml     context="ACCOUNTS"
    ├── 065-create-deferred-tables.xml   context="ACCOUNTS"
    └── 066-seed-coa-defaults.xml        context="ACCOUNTS"
                │
                ▼
Accounting tables created ONLY in this tenant's database
```

16.2 ServiceModule Enum Entry

```
public enum ServiceModule {
    // ... existing modules
    ACCOUNTS(false, "com.example.multi_tanent.accounts.entity"),
    // Dependencies: ACCOUNTS works best with PURCHASES and SALES
    // but can run standalone as a GL-only module
}
```

16.3 Module Dependency Check

```
When posting a PURCHASE INVOICE to GL:
    Accounts module checks:
        → Is PURCHASES module active for this tenant?
        → Proceed with AP posting

When posting a SALES INVOICE to GL:
    Accounts module checks:
        → Is SALES module active for this tenant?
        → Proceed with AR posting

When running WAREHOUSE GL entries:
    Accounts module checks:
        → Is WAREHOUSE module active for this tenant?
        → Proceed with inventory accounting
```

---

17. Security & Access Control

17.1 Accounting-Specific Roles

Role	Access Level
`ACCOUNTS_ADMIN`	Full access: COA, journal entries, reports, settings
`ACCOUNTS_USER`	Can create invoices, view reports, cannot modify COA
`ACCOUNTS_VIEWER`	Read-only: reports and statements only
`ADMIN`	Full access (platform admin)

17.2 Sensitive Operations

```
// COA modifications — ADMIN only
@PreAuthorize("hasRole('ACCOUNTS_ADMIN') or hasRole('ADMIN')")
public ResponseEntity<?> createGLAccount(...)

// Journal posting — cannot be undone without a reversal
@PreAuthorize("hasRole('ACCOUNTS_USER') or hasRole('ACCOUNTS_ADMIN')")
public ResponseEntity<?> postJournalEntry(...)

// VAT Return filing — ADMIN only
@PreAuthorize("hasRole('ACCOUNTS_ADMIN')")
public ResponseEntity<?> fileVatReturn(...)
```

17.3 Audit Trail

Every GL transaction is logged in `audit_log`:

```
audit_log:
    ├── action: CREATE / POST / REVERSE / MODIFY
    ├── entity: JournalEntry / Invoice / VATTransaction
    ├── entityId
    ├── changedBy (username)
    ├── changedAt (timestamp)
    └── previousValue / newValue (JSON snapshot)
```

Summary: How the Accounting Module Connects Everything

```
┌──────────────────────────────────────────────────────────────────┐
│                    KEY INTEGRATION POINTS                        │
│                                                                  │
│  Every business event in any module that has financial impact    │
│  MUST call the accounts_service layer to:                        │
│                                                                  │
│  1. accounts_service.postDocument(documentId, documentType)      │
│         → Generates JournalEntry + Lines                         │
│         → Saves to journal_entries table                         │
│         → Updates GL balances                                    │
│                                                                  │
│  2. The GL Engine uses ItemGLAccount + CompanyGLAccount          │
│     to determine WHICH accounts to debit and credit              │
│                                                                  │
│  3. VAT is automatically calculated and posted to                │
│     Current VAT Receivable / Current VAT Payable accounts        │
│                                                                  │
│  4. All postings are reflected IMMEDIATELY in:                   │
│     ├── Trial Balance                                            │
│     ├── General Ledger                                           │
│     ├── Income Statement                                         │
│     ├── Balance Sheet                                            │
│     └── Cash Flow Statement                                      │
│                                                                  │
│  5. The GL is the SINGLE SOURCE OF TRUTH for all financials.     │
│     No module stores financial totals — they all feed the GL.    │
└──────────────────────────────────────────────────────────────────┘
```

Document generated from: `Accounting Module (3).docx` + GTR Service Architecture Analysis

Scope: GTR Service ERP — Multi-Tenant SaaS Platform

Liquibase Context Tag: `ACCOUNTS`